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What's in a *t*

*Looking for the Afroasiatic verbal morpheme *t in Ancient Egyptian*

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Abstract

This paper explores potential connections between the Afroasiatic verbal morpheme *t and specific features of Ancient Egyptian verbal morphology. It focuses on three key cases: the passivizing morpheme t(ỉ/w), the suffix -t in the sḏm-t=f form, and the -t ending found in certain infinitives. Although all three have been previously linked to Afroasiatic *t, their functions and historical development remain uncertain. The study reexamines these proposals in light of recent scholarship. Particular attention is devoted to the infinitival -t, whose distribution is reassessed through an analysis of its associations with verbs across different semantic domains known to interact with *t in other Afroasiatic languages. Given the ambiguity of the Egyptian writing system and the limitations of the available data, the conclusions remain tentative. Overall, the paper emphasizes both the plausibility of these connections and the methodological challenges involved.

Keywords

Egyptian – passive t(ỉ/w) – sḏm-t=f form – infinitive t

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1 The Afroasiatic verbal morpheme **t* and Ancient Egyptian: Introduction

Ancient Egyptian verbs can be modified by several verbal morphemes that have clear parallels across the Afroasiatic language family. The most well-known of these is the causative morpheme *s-*, which remains fairly common in Middle Egyptian but becomes rarer and increasingly lexicalized in the later stages of the language (Müller, Matthias forthcoming; Schenkel 1999; Štubňová 2019; Kilani 2024).

Basic form		***s*- derived stem (causative)**	
mn	'to be established'	*smn*	'to establish'
ꜥnḫ	'to live'	*sꜥnḫ*	'to cause to live'

The morpheme *n-* is also attested, potentially with a detransitivizing function, though it is already rare and likely lexicalized in Old Egyptian (Štubňová Nigrelli 2022).

Basic form		**n- derived stem (intransitive)**	
kꜣ(*i*)	'to think about'	*nkꜣ*	'to ponder (intr.)'
ḫb(*i*)	'to reduce, decrease'	*nḫbḫb*	'to be smaller'

In contrast, the Afroasiatic **t* morpheme has no clear, universally accepted parallel in the derivative morphology of Egyptian verbs. Yet, Egyptian does present some *t* morphemes that might be connected to the Afroasiatic **t*. In this contribution, I will discuss what I consider to be the most promising candidates, namely the passivizing morpheme *t*(*i*/*w*), the *-t* of the so-called *sḏm-t=f* form and the *-t* ending found in some infinitives.

It must be stressed that this contribution is exploratory in nature and deliberately adopts a maximalist approach. The Egyptian morphemes discussed here should not be regarded as proven or securely established reflexes of the Afroasiatic **t* morpheme. Rather, the aim is to draw attention to cases that, from a possibilistic perspective, display features *potentially* compatible with an origin in that proto-morpheme.

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It is also worth noting that all three Egyptian morphemes considered here have already been tentatively linked by earlier scholars to the Afroasiatic morpheme, albeit with varying degrees of acceptance in the literature. In particular, Loprieno proposed as early as 1986 a common origin for these three Egyptian morphemes, connecting them to the Semitic and, by extension, Afroasiatic **t* morpheme, which he interpreted as originally perfective in value (see below).

Accordingly, this paper does not advance entirely new proposals concerning possible remnants of the Afroasiatic **t* morpheme in Egyptian. It is nevertheless not intended as a mere summary of previous scholarship either: while it agrees with Loprieno on the possibility that these Egyptian morphemes derive from a common Afroasiatic ancestor, it diverges from his interpretation of the morpheme's original function and, consequently, from his account of the pathways through which it was realized in Egyptian.

More generally, the main earlier proposals are here recontextualized in light of recent research, including several contributions collected in this volume. Whether the evidence provided is sufficient to support this hypothesis is left for the reader to determine. Personally, I do not claim certainty about any of these cases; however, I believe there is enough ground to at least entertain the possibility of such a connection.

It is important to emphasize that the speculative nature of this paper is not merely a stylistic choice, but rather an unavoidable consequence of the Egyptian evidence itself. The data is often frustratingly ambiguous and open to multiple interpretations, largely due to the inherent limitations of pre-Coptic scripts, which only recorded consonants and omitted vowels (and, with them, much of the grammatical information that would have clarified meaning). These issues are particularly relevant for the third case discussed here, the infinitive *-t* suffix, and will be explored in greater detail there. It is therefore clear that, at least in this domain, the insights provided by ancient Egyptian are inherently limited, and it is unlikely that the Egyptian data will significantly advance our understanding of the nature or function of this morpheme within the broader Afroasiatic family. Consequently, this paper can also be viewed as a methodological contribution, aiming to highlight the unique challenges, complexities, and limitations inherent in the Egyptian data; issues that are rarely addressed within Egyptological scholarship, and even more rarely outside it.

2 The passivizing morpheme *t*(*i̯*/*w*)

Egyptian possessed a morpheme *t*(*i̯*) (later *t*(*w*)) that, when added to an inflected verbal form, appeared to render it passive or impersonal. Syntacti-

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cally, this morpheme occurred after the verbal base and before the subject, if the latter was present. This morpheme is well attested in Old and Middle Egyptian, while it becomes increasingly rare in Late Egyptian and is finally lost in Coptic.

Various scholars have proposed that this Egyptian passive/passivizing morpheme may descend from this Afroasiatic **t* morpheme, although through different analytical paths. As mentioned above, Loprieno proposed this connection in his book on the Egyptian and Semitic verb (1986). In his view, the Egyptian *t*(*i*/*w*) resulted from the specialization of an earlier Afroasiatic morpheme **t* that was originally primarily a perfective marker, combined with diathesis-neutral verbal forms (1986, 45–46, note 62, 49–50). This idea stems from the observation that passive constructions with the suffix -t often appear to convey a perfective meaning, a feature that is absent from other passive forms lacking -t. This observation has also been highlighted more recently by Almansa-Villatoro (2024, 11–12) who proposes interpreting the -t suffix as a marker of prior circumstances.

This perfective > passive interpretation, however, does not seem to have gained favor in the field, even among scholars who agree with Loprieno on the likely Afroasiatic origin of this Egyptian suffix. In particular, more recently, Stauder (2023) defended the Afroasiatic origin of this Egyptian morpheme, but suggested conceptualizing its development as proceeding along what he defines as 'well-paralleled path leading from reflexive to anticausative to passive' (Stauder 2023, 185, 222–227). It is important to note that Stauder's article was written in response to the work of Díaz Hernández (2021, 2022) who instead suggested that the function of Egyptian *t*(*i*/*w*) as a passivizing verbal morpheme should be seen as a secondary development, resulting from the grammaticalization as an indefinite pronoun of what was originally a noun meaning 'body', 'person'.[1]

My own position on the matter is somewhat a middle ground. First, I agree with Loprieno and Stauder that Egyptian *t*(*i*/*w*) very likely ultimately derives from the Afroasiatic morpheme **t*. At the same time, I am more inclined to follow Stauder's proposed developmental path from an originally middle or reflexive morpheme to a passive one, rather than Loprieno's suggestion of a perfective origin. While Loprieno's idea had considerable merit given the Egyptian and Semitic data available to him at the time of publication, I believe

1 The idea that *t*(*i*/*w*) was in fact an indefinite pronoun predates Díaz Hernández' paper and can be found already in the works of earlier scholars (e.g. Westendorf 1953; Edel 1955; Grandet and Mathieu 1997).

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that more recent scholarship, drawing on a broader Afroasiatic perspective (including Cushitic) and on more recent analyses of the origins and functions of the Akkadian *-t-* morpheme, starting with Tanza-Kamil in this volume, points in a different direction. That said, I find Stauder's view that the *t*(*i̯*/*w*) attested in historical Egyptian should be interpreted primarily as a verbal suffix, with its use as an indefinite pronoun treated as a derived, secondary development, to be open to question, as it raises, in my opinion, important challenges.

In particular, it is true that in most cases the morpheme *t*(*i̯*/*w*) appears between the verb and the subject, with the subject taking the form of a suffix pronoun, as would indeed be expected if *t*(*i̯*/*w*) functioned as a true verbal affix (ex. 1).

(1) *šn=t=ṯ* *ḏs=ṯ* *nn* *srqt*
encircle= *ti̯/w_morpheme*=2SG.F self=2SG.F here scorpion
'You shall be encircled here, scorpion!' (PT 234a)

However, there is at least one instance already in Old Egyptian,[2] where *t*(*i̯*/*w*) can occur without any explicit subject following it (ex. 2), something that is usually not possible with other verbal suffixes, such as the prospective suffix *-w*.

(2) *ms~n=t*(*i̯*/*w*) *n*(=*i̯*)
give_birth~PST= *ti̯/w_morpheme* for(=1SG)
'Someone has given birth for me / It is for me that someone has given birth.' (PT 179a)

Stauder is aware of this form (2023, 202), but interprets it as nominal in nature, and thus inherently lacking an explicit subject: so *ms~n=t*(*i̯*/*w*) '(the act of) hav-

2 In the Pyramid Texts, so not only from the in the Twelfth Dynasty onward as suggested by Sauder (Stauder 2023, 227–228 and *passim*). Moreover, even if these forms were indeed first attested only in the Twelfth Dynasty, this would not imply that they originated at that time. One should therefore be cautious in assuming a direct diachronic development from the written language of the Old Kingdom to that of the Middle Kingdom, as has often been suggested. In my view, it is quite likely that these two written traditions instead reflect different dialects, that became prominent in writing at different point in time (Winand 2016, 2022). As a result, one could reasonably assume that the uses attested from the Twelfth Dynasty onward were already present in earlier stages of Egyptian and may even predate what is attested in Old Egyptian, but were confined to dialects that lay outside the written record.

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ing been born' = '(the) birth'. While I agree that this is in principle possible, I find the analysis somewhat strained; interpreting *tỉ/w* as an impersonal pronoun seems to me more plausible.

Furthermore, there is another problem with Stauder's argument. In example 2, the morpheme *t*(*ỉ*/*w*) is not only used without an explicit agent but also appears after the verbal suffix *~n*; this is not an isolated instance. This fact is significant, since the morpheme *~n* is generally regarded as originating from either a preposition or a particle of some kind (see Stauder 2014, 95–107 for a recent discussion). In such a position, one would expect only a noun or pronoun, and certainly not a true verbal suffix.

At the same time, these data do not necessarily invalidate Stauder's proposal of a degrammaticalization process from a passive suffix to an indefinite pronoun. Rather, they would simply imply that, if such a development did occur, it must predate Old Egyptian as attested in the Pyramid Texts and thus fall earlier than the period envisaged by Stauder, possibly even in prehistory.

Finally, it is also worth noting that the Egyptian *s-* and *n-* morphemes, which continue the Afroasiatic causative **s* and detransitivizing(?) **n* morphemes, are always placed before the verb in Egyptian. By contrast, the Egyptian morpheme *t*(*ỉ*/*w*) consistently follows the verb. This fact alone does not mean that *t*(*ỉ*/*w*) could not be a genuine verbal morpheme, since Egyptian does have verbal suffixes (e.g., the *-w* associated with the prospective). However, it does suggest an underlying difference between the etymological nature of the *s-* and *n-*, which emerged in Egyptian as prefixes, and that of *t*(*ỉ*/*w*), which appears as postponed morpheme. Whether this difference represents an Egyptian innovation or instead reflects a distinction already present at some pre-Egyptian stage is impossible to determine.

All this considered, the connection between the Egyptian *t*(*ỉ*/*w*) and the Afroasiatic **t* morpheme is possible, but clearly not as straightforward as in the case of the morphemes *s-* and *n-*.

3 The -t ending in the so-called *sḏm-t=f* form

Old and Middle Egyptian featured a verbal form traditionally referred to as the *sḏm-t=f* form, characterized by the suffix *-t* appended to the primary verb. The scope of its usage remains a subject of debate (see Zonhoven 1997; Brose 2024 for detailed discussions). The key issue is that many Middle Egyptian examples identified as *sḏm-t=f* forms appear to occur exclusively with verbs whose infinitives also featured the suffix *-t*. These infinitives are nearly indistinguishable from the *sḏm-t=f* form (see below), leading some to suggest that these

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instances are merely infinitives rather than true *sḏm-t=f* forms (Zonhoven 1997, 4–6 and *passim*).

Despite these ambiguities, a small set of three constructions is widely accepted as containing genuine *sḏm-t=f* forms, particularly because all types of verbs exhibit a final *-t* , even those that do not have it in their Middle Egyptian infinitives.

These three uncontroversial uses of the *sḏm-t=f* are the following:

a. In the negative pattern *n sḏm-t=f* , where it is used to express that an action or event has not taken place yet, i.e., before an action or event has taken place:

(3) *ḫpr rn=k n msy-t r(m)ṯ*
happen.PST name=2SG.M not be_born-*t*_SUFFIX people
'Your name happened when people had not been born yet / before people were born' (CT II 400a)

b. After the preposition *r*: 'until'

(4) Look, you are to spend month upon month …
r km-t =k ỉbd 4
until be_born-*t*_SUFFIX =2SG.M month 4
'… until you will have spent 4 months' (Shipwrecked Sailor 117–119)

c. After the preposition *ḏr*: 'before (temporal)'

(5) "Say my name", says the ground …
ḏr ḫnd-t =k ḥr =ỉ
before step-*t*_SUFFIX =2SG.M on =1SG
'… before you step on me' (CT V, 186f–g)

These examples show that the primary function of the *sḏm.t=f* form appears to have been to express the eventuality of a completed action in contexts where the action was not yet completed. This form can thus be described as both prospective and perfective, functioning akin to a future perfect tense. At the same time, this *sḏm.t=f* form is typically active and lacks any specific passive or middle connotation.

Several scholars have discussed the possible connections and developments of this form within Egyptian (see Brose 2024 for a recent overview), whereas the question of the pre-Egyptian origins, if any, of this *-t* suffix has attracted much less attention.

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The main proposal originates with Loprieno (1986, 124–129, *passim*) who compared the Egyptian form with the Akkadian *iptaras*, traditionally interpreted as a perfect, and argued that both derive from a common Egyptian–Semitic perfect proto-morpheme **t*. Within this framework, the perfect value attributed to both the Akkadian and the Egyptian forms would not constitute an innovation, but rather the retention of an original feature. This interpretation was subsequently adopted by Schenkel in his teaching material (2012) and more recently by Brose (2024, 53 and *passim*).

This connection, however, has also been explicitly rejected by Stauder (2014, 69), who notes that current scholarship on the Semitic side increasingly views the Akkadian "perfect" *iptaras* as an Akkadian-specific innovation. As such, it cannot be projected back to the Proto-Semitic level and therefore cannot be directly linked to the Egyptian form.

Here, too, my position occupies a middle ground. I agree with Stauder that current research favors an internal origin for the Akkadian *iptaras*, which may in fact not even constitute a true perfect (see Tanza-Kamil, in this volume, who argues that the form is better understood as a progression marker). More generally, as noted above, the evidence discussed in this volume strongly points to an originally middle or reflexive value of the Afroasiatic morpheme **t*.

Nevertheless, the fact that a shared perfect value can no longer be maintained, *pace* Loprieno, does not entail that the Egyptian and Akkadian *-t-* morphemes are unrelated or cannot ultimately be traced back to the same Afroasiatic **t*. In particular, I see no compelling reason to exclude the possibility that the Afroasiatic morpheme **t* underwent independent but parallel developments in Egyptian and Akkadian, resulting in a perfect or progression-marker function in both languages. If Akkadian underwent such a development, then in principle there is no reason, under a possibilistic approach, to exclude a comparable development in Egyptian.

4 The *-t* ending in Egyptian infinitives

Some Egyptian infinitives exhibit a final *-t* of uncertain origin and function. This suffix has received relatively little attention in Egyptological literature and is often implicitly regarded in standard grammars as a simple, semantically void verbal class marker. However, some of its properties and distributional patterns suggest that it may originally have been a meaningful morpheme, even if it became lexicalized in later phases of the language.

As already suggested by Loprieno (1986), it is reasonable to ask whether this Egyptian infinitival *-t* morpheme might be connected in some way to the

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Afroasiatic **t*. Therefore, the following sections of this paper address this *-t* ending in Egyptian infinitives.

I begin with a critical reassessment of the most common claim concerning this suffix, namely that it derives from a feminine marker. As already argued by Loprieno, there is clear evidence to reject this interpretation, as well as the closely related assumption that the suffix *-t* functions merely as a semantically empty class marker.

I then discuss the two existing proposals that attempt to connect this morpheme with the Afroasiatic **t*, namely Loprieno's proposal and my own recent one. Finally, I present new quantitative data on the distribution of this suffix across verbs belonging to specific semantic domains that are often associated with the **t* morpheme in other Afroasiatic languages, which, in my view, further strengthens the case for my interpretation.

4.1 *The infinitive* -t*: A feminine ending?*

The infinitive *-t* has often been described as a 'feminine ending' (which thus produced 'feminine infinitives') in Egyptological literature, though this characterization is typically made without any explanation as to why the ending is interpreted as feminine (Edel 1955, 344–351; Gardiner 1957, 223; Gundacker 2013).

Two considerations seem to underlie this analysis. First, there is the superficial similarity between this infinitival *-t* and the feminine *-t* suffix. Second, already in Old Egyptian there are instances of forms that have been interpreted as action nouns, end in *-t*, and are treated as grammatically feminine, as indicated by agreement with other elements in the clause (see ex. 6).

(6) *pr-t* *=k* *tn* *m* *pr* *=k*
go_out-*t*_SUFFIX =2SG.M this.F from house =2SG.M
'This going out of yours from your house' (PT 579.1)

Here the verbal form *pr-t* can be identified as nominal due to its specification by the deictic *tn*, which exclusively specifies nominal elements. It can also be assumed to be feminine, as the deictic *tn*, which must agree in gender and number with the noun it modifies, is unambiguously feminine. Yet, it is important to note that instances of forms that *could* be interpreted as infinitives whose gender is made explicit by the agreement of other elements in the sentence are exceedingly rare in Old Egyptian. Only a handful of examples exist in the Pyramid Texts, and they all appear to be limited to the verbs *pr*(*ỉ*) : 'to go out' and *šm*(*ỉ*) : 'to go' (e.g., PT 436.7). In all these (very few) cases, the gender does indeed appear to be feminine.

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Things are significantly different in later phases of the language, not only because the evidence about the gender of action nouns/infinitives is more abundant, but also because (as already noticed by e.g. Loprieno 1986, 49–50, note 72) such evidence clearly points to a masculine gender for these forms.

The strongest argument in this regard comes from Late Egyptian (attested from the middle of the second millennium BCE) and later stages of the language. Unlike Old and Middle Egyptian, where definiteness was not independently marked, Late Egyptian and its successors introduced articles that agreed in gender and number with their associated nouns, making this distinction explicit.

Now, in the case of infinitives characterized by a final *-t*, the article used is typically the masculine *pꜣ*, not the feminine *tꜣ* (ex. 7).

(7) *pꜣ* *ỉn-t* *=k* *r* *sḫt* *=n*
ART.SG.M to_bring.INF-*t*_SUFFIX =2SG.M in_order_to to_beat.INF =1PL
'the act of bringing (infinitive) you in order to beat us' (Papyrus BM 10247 18,2)

This clearly indicates that in Late Egyptian, these infinitives were perceived as grammatically masculine despite their final *-t*. This analysis is further supported by nouns that can reasonably be assumed to derive from lexicalized action nouns/infinitives in *-t*, and which appear to be grammatically masculine in Late Egyptian, Coptic, or both. A good example is provided by the Coptic word [S][3] ⲉⲓⲃⲉ |íbə|[4] : 'thirst'/ 'to be thirsty'|, attested in Sahidic and other dialects, which is clearly derived from the pre-Coptic action noun/infinitive *ỉb-t* (already attested with the *-t* suffix in Old Egyptian with the same meaning, see e.g., PT 211.2). In Coptic, this form functions both as an infinitive meaning 'to be thirsty' and as a grammatically masculine noun meaning 'thirst' (Crum 1939, 76b).

How can we reconcile this contrasting evidence, which suggests a feminine gender in the Old Egyptian examples discussed above, but points to a masculine gender in these later forms? I see two possible explanations:

1. One possible scenario is that the action nouns and infinitives in *-t*, originally feminine in Old Egyptian, came to be reinterpreted as masculine in later phases of the language or in the dialects underlying them. While

3 I use [ ] to indicate the specific Coptic dialect the form comes from, using the common abbreviations (S = Sahidic, etc.).

4 I use || to indicate the phonological interpretation/reconstruction of the Coptic form (see Kilani 2025, 123 for discussion).

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such a shift is not impossible in principle (for example, it could have resulted from leveling caused by the greater number of masculine infinitives without *-t*) it seems unlikely to me. Not only would this have been a fairly extensive change, but there is also nothing in the forms themselves that could have suggest a masculine reinterpretation. On the contrary, in Coptic and likely in Late Egyptian, the final unstressed -ϵ that typically marks these forms is generally associated with feminine nouns rather than masculine ones, yet the forms appear to have been treated as masculine.

2. A second possible scenario is that the action nouns/infinitives in *-t* were masculine in both Old Egyptian and later stages, and that the few forms that appear to be feminine in Old Egyptian are not true action nouns/infinitives. For instance, while an action nouns/infinitives (e.g., "the act of going out," which might be conceptualized as [+processual]) could fit the context, an abstract noun (e.g., "the exit" or "the departure," which can be conceived as reified and thus [-processual]) is also possible. In this respect, it is important to note that Egyptian did have derived feminine abstract nouns.[5] This possibility deserves attention, since our interpretation of these forms depends only on *expected* sentence meanings and *assumed* grammatical structures, yet the inherent ambiguity of the vowel-less hieroglyphic script leaves abundant room for interpretation. Moreover, the Pyramid Texts are collections of magical and ritual formulae whose nuances are often difficult to understand. Last but not least, the fact that the only clear examples come from two closely related motion verbs might be due to the specific semantics of those verbs rather than a general pattern.

To sum up, while a few rare clearly feminine forms that might be interpreted as infinitives occur in Early Egyptian, later evidence clearly shows that infinitives in *-t* were actually masculine. In this light, although a gender shift cannot be entirely ruled out, it seems simpler to assume that the *-t* of infinitives was unrelated to the feminine marker *-t*, and that the few apparently feminine forms in Old Egyptian were not true infinitives to begin with. This, in turn, implies the existence of a *-t* morpheme associated with infinitives that was not a feminine marker.

5 E.g., *nfr-t* : 'beauty' < √*nfr* : '(to) be(come) beautiful', *ḏw-t* : 'evil' < √*ḏw* : '(to) be(come) evil', *mrw-t* : 'love' < √*mr̆* : '(to) love', *msw-t* : 'birth' < √*mš* : '(to) give birth', all feminine.

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4.2 *The Egyptian suffix* -t*: What was its original function?*

The traditional identification of *-t* with the feminine ending carries another implicit assumption, namely that the presence or absence of this morpheme was simply a feature associated with certain verbal classes: some classes had masculine infinitives, while others feminine ones, without any specific reason for this distribution. This reasoning, which dominates Egyptological grammars (see e.g. the grammars of Gardiner 1957; Grandet and Mathieu 1997; Allen 2014, etc.), is flawed in two respects. First, as discussed above, there is strong evidence that the *-t* of infinitives was not a feminine after all. Consequently, if this morpheme was not merely a feminine marker, it is reasonable to assume that it must have served some other grammatical function at some point in the history of the language. Second, clear evidence from both Egyptian and Coptic shows that verbs traditionally assigned to classes that were supposedly characterized by infinitives in *-t* could also appear with infinitives lacking *-t*; this undermines the idea that *-t* functioned solely as a class marker. In fact, as will be discussed in more detail below, the very definition of traditional Egyptological verbal classes is problematic and has recently been called into question.

These observations raise two interconnected questions. The first point concerns the original function of the Egyptian infinitival *-t* morpheme, while the second concerns the reasons for its compatibility with the infinitives of certain verbs but not others. To my knowledge, only two attempts have addressed these questions. The first, by Loprieno, considers only the first question and offers no explanation for the second. The second, briefly introduced by the present author in a recent publication, proposes a framework that addresses both. Both solutions explain the *-t* suffix of Egyptian infinitives as derived from the Afroasiatic morpheme **t*, but they do so in different ways.

4.3 *The Egyptian suffix* -t *in infinitives according to Loprieno 1986*

Loprieno approaches the issue of infinitives in *-t* in his previously mentioned book on the Egyptian and Semitic verbal system. His approach aligns with his broader reasoning and with what has already been discussed in the previous paragraphs: he assumes a primarily perfective meaning for the proto-morpheme **t* and suggests that most, if not all, non-feminine *-t* affixes appearing in verbal forms are ultimately derived from it. It logically follows that the *-t* suffix of infinitives must also be a manifestation of this perfective proto-morpheme **t*. In fact, he proposes that the nominal form later emerging as the infinitive in the historical language, is etymologically the starting point for the other verbal forms in *-t* like the *sḏm-t=f* discussed above (Loprieno 1986, 44–45, 49–50, and *passim*).

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While intriguing, Loprieno's solution presents two problems. The first is the one highlighted above: although a perfective value for the proto-morpheme might have seemed reasonable at the time the book was published based on Egyptian and Semitic evidence alone, it is hardly defensible today in light of more recent scholarship and a broader Afroasiatic perspective. A further problem with Loprieno's hypothesis is that, while he argues the original perfect nominal form in *-t* surfaces as the infinitive of *certain* verbal classes, he does not explain *why* it appears *only* with those verbs. In other words, his hypothesis does not (and in my opinion cannot) account for how this supposedly perfect nominal form became an infinitive *for some* verbs while disappearing entirely *for others*, leaving the second question raised above unaddressed.

4.4 *The Egyptian suffix* -t *in infinitives according to Kilani 2025*

In contrast to Loprieno's hypothesis, the proposal recently advanced by the present author (Kilani 2025) not only aligns more closely with current views on the Afroasiatic morpheme *-*t*, but also offers a framework for addressing both questions raised above. The issue of the *-t* in infinitives, however, was not the primary topic of that study; rather, it arose naturally from the critical reassessment of so-called Egyptian verbal classes that constituted its main focus. This reassessment was prompted by a simple observation: although traditional Egyptological verbal classes have been in use for over a century, their definitions are clearly problematic, both internally and in comparison with the Coptic evidence. The study thus proposed a revised definition of these classes and a reclassification of Egyptian-Coptic infinitives. Because the new proposal for interpreting the Egyptian suffix *-t* directly stems from these considerations and from the framework developed in that study, it is necessary to summarize its key points here.

4.4.1 The traditional verbal classes and their internal contradictions (Kilani 2025, 124–127)

Egyptian verbal roots are traditionally divided into 'verbal classes' based on their consonantal structures across various verbal forms. While these 'classes' might superficially appear (and are often mistakenly assumed) to categorize verbs *only* on the basis of the structure of their consonantal roots (as it is the case for root classes in Semitic studies), their actual nature is far more complex and idiosyncratic (see Gardiner 1957, 206–213).

In few words, this system categorizes verbs on the basis of three main factors, namely:

1. Paradigmatic alternations across different morphological forms (e.g., presence or absence of reduplication in selected verbal forms, presence or absence of the *-t* ending in the infinitive)

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2. Number and types of root consonants as appearing in writing at specific points in time (both proper root consonants, but also the causative prefix *s-*)
3. Inclusion of certain hypothetical root consonants that may not be attested in Middle Egyptian, while other consonants unattested in Middle Egyptian but present in Coptic or other Afroasiatic languages are excluded (this applies especially to weak consonants or glides *-ỉ*/*-w*, which can either be ignored despite Afroasiatic comparative evidence, or assumed even in the absence of direct supporting evidence)

It is worth noticing that the presence or absence of the extra *-t* in the infinitive is a key factor in classifying a verb into one class or another; however, this approach involves a somewhat circular reasoning (see below).

Leaving aside a couple of irregular verbs, the combination of these factors results in the following 'verbal classes':

– **2-lit, 3-lit, 4-lit, 5-lit verbs**: non-causative verbs whose basic form is characterized by 2, 3, 4 or 5 consonants, whose last consonant is not a weak consonant (*-ỉ* and *-w*). Some of these verbs can display reduplication of the consonant preceding the final week one in some verbal forms (e.g., some participles), but not in others. **4-lit** and **5-lit** verbs are mostly derived verbs built through the partial or full reduplication of the root (usually as 1212, 12323, or 1313 in case of II-ꜣ roots)
– **III-weak, IV-weak verbs**[6]: non-causative verbs whose basic form is characterized by 3, or 4 consonants, whose last consonant is attested or assumed to have been a weak consonant (*-ỉ* and *-w*). Some of these verbs can display reduplication of the consonant preceding the final week one in some verbal forms (e.g., some participles), but not in others.
– **II-gem and III-gem**: non-causative verbs whose basic form is characterized by 2, 3 consonants, and which display widespread reduplication of the last consonant also in verbal forms where such reduplication does not happen in other classes.
– **Caus-2-lit, etc., Caus-II-weak, etc., Caus-II-gem, etc.**: *s-* prefixed causatives of the previous classes.

This classification presents several conceptual issues (see Kilani 2025 for a detailed discussion). First, it conflates features belonging to different grammatical levels: on the one hand, it groups together structural properties of the root, such as the number of consonants—a criterion that is indeed central to Semitic root classification; on the other, it incorporates additional characteristics such

6 Or III-inf, IV-inf from Latin 'infirmae'.

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as the presence or absence of extra root morphemes, which are more appropriately understood as phenomena operating at the level of the stem (in the Semitic sense of the term).[7] Moreover, it also relies on inflectional behaviors, such as reduplication in certain verbal forms, which are better interpreted as paradigmatic rather than classificatory features. Moreover, this classification is largely (though not exclusively) based on the forms verbs take in written Egyptian, which is inherently problematic given the defective nature of the script. Egyptian writing does not indicate vowels or geminate consonants[8] and is often ambiguous even in its representation of regular consonants, especially weak ones. In addition, many verbs are attested only in a limited number of forms rather than across their full paradigms, which further complicates any attempt at a reliable classification.

All this results in a highly idiosyncratic system, which is more a projection of various superficial observations and long-standing Egyptological assumptions than an objective representation or even an approximation of an actual linguistic reality.

4.4.2 The traditional verbal classes and their incompatibility with the Coptic evidence (Kilani 2025, 141–161)

One of the central assumptions of the traditional class system is that each class is characterized by a single type of infinitive (e.g., with or without *-t*), which is therefore treated as diagnostic for identifying the class of the verbs that display it (see e.g. Grandet and Mathieu 1997, 171–174; Allen 2014, 164). The only traditionally agreed-upon exceptions are verbs interpreted as so-called IV-weak, which are generally recognized as having infinitives both with and without *-t*. Other cases of infinitives that do not fit the traditional pattern are usually mentioned only in passing, without further consideration of the implications of their existence (e.g. Gardiner 1957, 224), or are dismissed as mistakes, exceptions, or phonological variants (e.g., Allen 2014, 165).

7 This is most clearly the case with causative verbs, which are often treated as independent verbal classes in Egyptological literature (e.g. Allen 2014, 157–158), despite the fact that they are transparently derived stems characterized by the prefix *s-*, rather than genuinely independent verbal roots. So-called IV-weak verbs have likewise been argued to represent a derived stem rather than a true verbal class (see, e.g., Lacau 1953, 26), since some appear to be based on independently attested triradical roots extended by an additional final vowel, weak consonant, or glide.

8 I use here and below the term geminate consonants in its common linguistic meaning, i.e. to refer to (usually intervocalic) long consonants (VC:V). This clarification is needed because in Egyptological literature the term "geminate" is unfortunately used mostly to refer to the reduplicated consonants likely separated by a vowel (CVC), rather than true geminate consonants.

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However, when Coptic is taken into consideration, it quickly becomes clear that the situation is more complex. In particular, not only do several Coptic verbs display multiple infinitives, but these distinct infinitives often take forms that would be considered diagnostic of different verbal classes under the traditional system. For instance, the verb *ḫn* : 'to approach, to become near' has in Coptic both an infinitive [S] ϩⲱⲛ |hón|, traditionally considered typical of 2-lit verbs, and an infinitive [Sa] ϩⲛⲁⲛ |hᵊnán|, traditionally associated with so-called II-gem verbs. Similarly, the verb *tk* : 'to throw' displays both an infinitive [A] ϯⲕⲉ |tíkə| < *tk-t*, usually assumed to be specific to so-called III-weak verbs, and an infinitive [A] ⲧⲱⲕ |tók| < *tk*, usually attributed to 2-lit verbs (Kilani 2025, 144, 152).

This multiplicity of infinitives, along with the fact that different verbs display different combinations of forms, suggests two important points. First, it indicates that specific infinitives are not uniquely tied to specific root types and therefore cannot serve as reliable diagnostic criteria for identifying verbal root classes. Second, it points to the existence of multiple verbal stems and vocalic patterns, each characterized by specific infinitives, through which the underlying root could be realized.

4.4.3 A revised system of verbal root classes and verbal nouns: Key elements

The observations discussed in the previous paragraphs formed the basis for the new class system proposed in Kilani 2025. It is not necessary to review all the details of that system here, and interested readers are referred to the main article. For the purposes of the present discussion, it is sufficient to highlight three key elements of the system, as they bear directly on the role of the *-t* morpheme. They are the following:

1. Egyptian verbal roots can be classified into two main categories, one of which appears to result from the merging of two subcategories. The first category, labelled *3-EtR*, comprises all roots characterized by three distinct strong consonants.[9] This category corresponds to the so-called *3-lit* and *IV-weak* classes of the traditional system. The second category, labelled *2⁺-EtR*, comprises all roots characterized by two distinct strong consonants. These roots likely had a third weak consonant, either as the second (*II-weak*) or the third (*III-weak*) element of the root. However, such weak consonants are usually not visible in writing at any stage of

9 Although not explicitly stated in the original article, roots with initial *ỉ-* or *w-* can also be included in this class, since these consonants behave as strong consonants in initial position.

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the language, and by the time of Coptic, these original *II-weak* or *III-weak* roots appear to have merged in most, if not all, contexts.[10] As a result, it is generally impossible to identify the exact form of the root, and both types are grouped under the common label *2⁺-EtR*. The *2⁺-EtR* class of my system encompasses roots traditionally assigned to the *2-lit*, *III-weak*, and *II-gem* categories.

2. Roots were realized through stems characterized by one-consonant prefixes (e.g., the causative *s*-)
3. Roots could also be expanded through one (or more?) suffix formed by either a vowel, or a vowel + glide. I conventionally transcribe it as *-Vy̆*.[11]

These three elements provide the scaffolding for the entire verbal system. Within this framework, several infinitives, or more precisely non-actant verbal nouns,[12] can be identified for each root. In addition to stem prefixes and *-Vy̆* suffixes, these non-actant verbal nouns are characterized by two further features, each structured as a binary opposition, namely:

1. An opposition between two contrastive vocalic patterns, namely:
   a. *V́-V* for action nouns—e.g., [S] ϣⲱⲧⲙ |šótᵊm| : 'to close (something)', 'the action of closing'
   b. *V-V́*[13] for state/result nouns—e.g., [S] ϣⲧⲟⲙ |šᵊtóm| : 'to be closed/closing', 'the state of being closed/closing'
2. A binary opposition between a Proto-Coptic vowel **a* and a non-**a* vowel, conventionally transcribed here as **E*. These two vowels appear to have encoded a grammatical contrast at the level of diathesis or agentivity. While the precise nature of this opposition remains to be clarified, the available evidence suggests that:
   a. The vowel **a* was associated with active agency or, in general absence of any external agent—e.g., [S] ϣⲧⲟⲙ |šᵊtóm| < **ḫVtám* : 'to be closed/closing', 'the state of being closed/closing'
   b. The vowel **E*, instead, seems to be associated with a non-actant meaning and with the presence of an implicit or explicit external

10 Since in most cases the exact position of this weak consonant cannot be established, and it can only be assumed to have occurred either before or after the second strong consonant of the root, I will transcribe it with a cross˟ placed over the second consonant of the root.

11 The function of this suffix was left as an open question in the original article. I am now inclined to think that it may have encoded some form of deontic, prospective, or volitional/intentional modality, though this hypothesis requires further investigation.

12 In opposition to actant verbal nouns which are more akin to participles.

13 Possibly from an earlier V-V́: , with a long vowel in the second syllable.

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agent—e.g., [S] ⲥⲛⲁⲧ |sᵊnát| < *sVnÉḏ : ‘to be afraid (by/because of something/someone)’, ‘the state of being afraid (because of something/someone)’

When these two features are combines, we obtain the following four basic patterns

	Ý-V **action noun**	*V-Ý* **state/result nouns**
**a* – external agent	*á-V*	*V-á*
**E* + external agent	*É-V*	*V-É*

4.4.4 The Egyptian suffix infinitival *-t* within the revised system (Kilani 2025, 157–158)

When the distribution of the suffix *-t* is examined within this new framework, two points become clear. First, no correlation can be observed between the presence of *-t* and any of the following factors: root class (*3-EtR* vs. *2⁺-EtR*), stem type (basic vs. s-causative), or the presence or absence of the *-Vỷ* suffix. Infinitives both with and without *-t* occur across all these categories. This demonstrates that *-t* does not function as a marker for any of these grammatical distinctions.

By contrast, when the distribution of the suffix *-t* is examined in relation to the two additional features that characterize verbal nouns, a clear correlation emerges. The *-t* suffix appears *only* in one of the four possible configurations, namely on verbal nouns that are a) stressed on the first syllable and b) containing the vowel **E*. This pattern is attested in the basic stem (e.g., [B] ϩⲉⲙⲥⲓ |hémsi| < **ḥÉmsV-t* : ‘to sit’), in verbs extended with the *-Vỷ* suffix (e.g., [S] ϣⲙϣⲉ |šémsə| < **šÉmsV-ỷ-t* : ‘to follow’, ‘to serve’, ‘to worship’), in s-causatives (e.g., [S] ⲥⲙⲓⲛⲉ |sᵊmínə| < **sVmÉn̆V-t* : ‘to remain’, ‘to be established’), and in s-causatives further extended with the *-Vỷ* suffix (e.g., [S] ⲥⲁϩⲛⲉ |sáhnə| < **sÉḥn̆V-ỷ-t* : ‘to provide’).[14]

14 For the s-causatives extended with the suffix *-Vỷ* see the paper currently under review. The very few s-causatives without *-Vỷ* will be discussed in a future paper.

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As noted above, verbal nouns stressed on the first syllable appear to be primarily action nouns, which fits well with their use as infinitives. The vowel **E*, by contrast, seems to be associated with the presence of an external actor. In the case of action nouns, this combination would therefore be expected to yield something akin to a passive or non-active infinitive. However, as the examples just mentioned show, the infinitives with *-t* tend to have active meanings. How, then, can this apparent mismatch be explained?

Starting from the assumption that the Egyptian suffix *-t* may indeed be related to the Afroasiatic morpheme **t* in its middle or reflexive function, and taking into account the often indefinite or impersonal interpretation associated with this suffix in Egyptian, I argued (2025, 157–158) that the vocalization in **E* and the ending *-(V)t* fulfilled complementary grammatical roles. Specifically, the vowel **E* would mark the main argument of the verb as a non-actant experiencer (gloss: NONACT) of an event initiated by an external agent, while the suffix **-(V)t* would indicate that this external agent is either undefined or unspecified (gloss: INDF/IMPERS), or identical with the main argument itself, yielding a reflexive interpretation (gloss: REFL).

This analysis would help explain why infinitives containing the non-actant vowel **E* can nonetheless display an apparently active meaning, because within this framework, an infinitive such as [S] ꙗıcɛ |mísə| < *ms-t*, traditionally analyzed as **mÉšV-t* and glossed as [birth-INF], meaning 'to birth' or 'the act of giving birth', could instead be interpreted as:

> [S] ꙗıcɛ |mísə| < **m⟨É⟩š-t* = [birth.INF⟨NONACT⟩-INDF/IMPERS]
> 'the act of being born by someone' / 'to be born by someone' → 'to give birth'

Similarly, a form such as *pr-t* < **pÉřV-t*, usually glossed as [go_forth-INF] and translated as 'to go forth' or 'the act of going forth', might instead be analyzed as

> [S] ııpɐ |pírə| < **p⟨É⟩ř-t* = [moving_forth.INF⟨NONACT⟩-REFL]
> 'to be moved forth by oneself' → 'to move (oneself) forth'.

As for the verbs mentioned above, I would thus suggest analyzing them as follow:

> [B] ƨɐʜcı |hémsi| < **ḥ⟨É⟩msV-t* = [be(coming)_seated.INF⟨NONACT⟩-REFL]:
> 'to be seated by oneself' → 'to sit'

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[S] ⲱⲙⲱⲉ |šémsə| < *š⟨É⟩*msV-ỷ-t* = [move_after.INF⟨NONACT⟩-?-REFL]:
'to be moved by oneself after (something/someone)' → 'to follow'

[S] ⲥⲁϩⲛⲉ |sáhnə| < *s⟨É⟩*ḥn̆V-ỷ-t* = [cause_to_take_care.INF⟨NONACT⟩-?-REFL]:
'to be caused by oneself to take care (of something/someone)' → 'to cause oneself to take care', 'to provide'

[S] ⲥⲙⲓⲛⲉ |sᵊmínə| < *s*Vm⟨É⟩*n̆V-t* = [cause_to_be_stable.INF⟨NONACT⟩-REFL]:
'to be caused by oneself to be stable' → 'to remain'

as well as:

[S] ⲥⲙⲓⲛⲉ |sᵊmínə| < *s*Vm⟨É⟩*n̆V-t* = [cause_to_be_stable.INF⟨NONACT⟩-INDF/IMPERS]:
'to be caused by someone to be stable' → 'to be established'

This interpretation would also seamlessly account for the fact that suffix pronouns attached to these infinitives can function either as the patient of the action with transitive verbs or as their agent with intransitive ones. Thus, a form such as [S] ⲙⲁⲥⲧϥ |mástᵊf| < *ms-t=f* : 'to birth him' or 'the birthing of him' could be interpreted as:

m⟨É⟩š-t=Vf = [birth.INF⟨NONACT⟩-INDF/IMPERS=3SG.M],

literally 'his being born by someone' or 'to have him born by someone'.

A form such as *pr-t=f* : 'to go forth of/by him' or 'his act of going forth', could instead be analyzed as

p⟨É⟩r̆-t=Vf = [moving_forth.INF⟨NONACT⟩-REFL=3SG.M]

to be understood as 'his being moved forth by himself' or 'to move himself forth'.

4.4.5 The distribution of the Egyptian infinitive -*t*and verbal semantics
The analysis presented in the 2025 article and summarized above was not only based on a reassessment of verbal classes and infinitives but was also informed by preliminary results presented at the workshop from which this volume originates. At the workshop, it was argued that Egyptian infinitives displaying the

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suffix *-t* are not evenly or randomly distributed across semantic fields, but instead tend to cluster in domains that, in other Afroasiatic languages, are frequently associated with mediopassive, reflexive, or autobenefactive meanings. This distribution is suggestively consistent with the hypothesis presented here: if the suffix *-t* does indeed derive from a middle or reflexive Proto-Afroasiatic morpheme **t*, then it is reasonable to expect it to combine preferentially with verbs that are semantically compatible with such nuances. Because the results presented at the workshop were preliminary, they were mentioned but not fully developed in the 2025 article; it is now possible to present them in greater detail.

4.4.5.1 *The limits of a qualitative approach*

One might begin by noting that, in principle, the most straightforward way to assess the semantic contexts in which the suffix *-t* occurs, and thus to infer its original function, would be to collect attested instances of action nouns or infinitives with and without *-t* in Old Egyptian, the earliest attested stage of the language, and to extrapolate their meanings from their contexts of use. In practice, however, the nature of the Egyptian evidence renders such an approach essentially impossible. On the one hand, it is often unclear whether a given form represents an action noun with a verbal *-t*, an abstract noun with the feminine *-t*, or some other type of derived forms altogether. On the other hand, our understanding of the relevant textual contexts is frequently too indeterminate, and the range of possible semantic nuances associated with the Afroasiatic morpheme **t* too broad and subtle to allow for secure conclusions. As a result, it would be very easy to argue for the presence of such a value in most, if not all, cases, were one inclined to do so.

One example will suffice to illustrate this issue. As already observed by Allen (1984, 451–452), the Pyramid Text spell PT 536 and PT 676 provide a very similar sentence in which a nominal form of the verb *ỉp* : 'to count', which should belong to a verbal class without *-t*, appears once with an extra *-t* (ex. 8) and once without it (ex. 9).

(8) PT 536 20–21

hrw	*pw*	…	*n*	*ỉp-t*	*qs-w*
day	this.M.SG		of	count.INF-*t*_SUFFIX	bone-PL

(9) PT 676.29

hrw	*pw*	*n*	*ỉp*	*qs-w*
day	this.M.SG	of	count.INF	bone-PL

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The form can be interpreted as infinitival in nature, and the general meaning of these sentences can be expressed as 'this day of counting (the) bones'. Based on this, Allen (1984, 451–452) interprets the *-t* in example 8 as a verbal morpheme with a meaning or function so subtle that it could be added or omitted without significantly affecting the overall meaning of the sentence.

Yet, which interpretation would make more sense here: a feminine suffix possibly implying an abstract noun or a reflex of the Afroasiatic **t*? Unfortunately, since there are no external elements to reveal the actual gender of these forms, either hypothesis could be supported. If one interprets it as a feminine suffix, the form with *-t* could be seen as more abstract, while the form without *-t* might appear more concrete—for instance, translating the first as 'the day of the enumeration (as an abstract reality) of the bones', while the second as 'the day of counting (as a specific action) the bones'. Conversely, if one wants to see it as an instance of the Afroasiatic **t*, the variant without *-t* might be interpreted as a plain sentence such as 'the day of counting the bones', while the variant with *-t* could be given a middle or autobenefactive nuance, translating it as 'the day of counting one's own bones' or 'the day of counting bones for one's own sake', or something similar. Given that this is a rather obscure religious reference, and we lack any clear understanding of what the 'day of *ỉp*(*t*) the bones' truly means, there is no objective basis to favor or exclude any of these interpretations.

It is thus evident that a qualitative analysis of each attestation will not help us determine the underlying function and semantic associations of this morpheme. Simply put, the consonantal nature of the script obscures part of the linguistic signal, the contexts in which these forms appear are often too vague, and numerous valid (albeit potentially ad hoc) explanations could be advanced in favor of one interpretation or another.

4.4.5.2 *The potential of a quantitative approach*

While a qualitative approach appears unworkable, a quantitative approach, though unlikely to resolve the issue conclusively, may nevertheless yield valuable insights. The underlying logic is straightforward. If the Egyptian suffix *-t* had no specific semantic value, or if it functioned merely as a generic feminine or abstract marker, then in principle it should show no correlation with particular semantic classes of verbs, since abstract nouns could be derived from any verbal root. Under this assumption, one would expect infinitives in *-t* to occur with roughly equal frequency across different semantic types of verbs, such as grooming verbs, verbs of motion, and so forth. This expectation constitutes our null hypothesis.

If, however, verbs attested with at least one infinitive in *-t* are found to cluster disproportionately in certain semantic categories, such distributions are

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unlikely to be accidental and may provide clues to the original function of the morpheme. Moreover, if the semantic categories that correlate with the presence of *-t* in Egyptian also correspond to those most commonly associated with the **t* morpheme in other Afroasiatic languages, then this would strongly suggest that the Egyptian infinitival *-t* is indeed ultimately related to that broader Afroasiatic morpheme.

Obviously, the fact that the Afroasiatic **t* morpheme can have so many generic values is an issue also for this approach, as in principle one could easily imagine ways (and find parallels in other Afroasiatic languages) in which any generic transitive or intransitive verb could have been compatible with it.

One way to mitigate this problem would be to first define more precisely the verbal semantic categories that are most likely to interact with the **t* morpheme across the Afroasiatic languages, and then to examine how verbs belonging to these t-compatible categories behave in Egyptian. To test this hypothesis, two elements are required: a corpus of Egyptian verbs for analysis, and a list of semantic fields likely to be associated with (and ideally diagnostic of) the Afroasiatic **t* morpheme.

For the verb corpus, I will use the verbs listed in the lexicon recently published at the end of Allen's edition of the Pyramid Texts of Unas (Allen 2017). This study will focus specifically on *2*$^{+}$*-EtR* without *-t* (*2*$^{+}$*-EtR* [–t]), *2*$^{+}$*-EtR* with *-t* (*2*$^{+}$*-EtR* [+t]), *3-EtR* without *-t* (*3-EtR* [–t]), *3-EtR* with *-t* (*3-EtR* [+t]) verbs, as other classes are largely marginal. The corpus is extensive and comprehensive, comprising 314 verbs. Moreover, because it is drawn from Old Egyptian, it reflects an early stage of the language—one that, in principle, is more likely to preserve the original meanings of these verbs and forms.

As for the semantic domains, regrettably, we currently lack a systematic, cross-Afroasiatic overview of the semantic range and compatibility of the Afroasiatic **t* morpheme. However, a few studies have been conducted on Akkadian and other Semitic languages (see e.g., Kouwenberg 2005; 2010; Notarius 2021; Sjörs 2018, 5–6, n. 19), as well as on the Cushitic language Somali (Saeed 1995), revealing a series of semantic categories that seem to be associated with the **t* morpheme in at least some Semitic and Cushitic languages. These categories include: RECIPROCAL ACTIONS, GROOMING VERBS AND OTHER REFLEXIVE ACTIONS AFFECTING THE SUBJECT'S BODY, GIVING BIRTH, EMOTION AND COGNITION VERBS, SELF TRANSLATIONAL AND POSTURAL MOVEMENTS, MOVEMENT OF THE OBJECT BY THE SUBJECT, UNCONTROLLED INCHOATIVE, and a broad, somewhat vague category described as AUTOBENEFACTIVE. While this framework is far from definitive, it provides a useful starting point for a preliminary assessment. However, before applying it to Egyptian verbs, a few adjustments are necessary.

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First, it is helpful to create some subcategories. In particular, I divide SELF TRANSLATIONAL AND POSTURAL MOVEMENTS into the following: basic movement (e.g., 'to come', 'to go'), movement form a source (e.g., 'to leave'), movement through something (e.g., 'to pass through'), type of movement (e.g., 'to walk', 'to run'), change of direction (e.g., 'to turn back'), movement toward/into/onto a target (e.g., 'to ascend'), movement around something (e.g., 'to go around'), moving from one place to another (e.g., 'to wander'), and postural movements (e.g., 'to sit'). Similarly, MOVEMENT INVOLVING OTHERS can be subdivided into actions such as to take/carry, to put/give/place, to send away, and to move apart/separate.

Moreover, uncontrolled inchoative verbs present challenges in Egyptian, as the same verbal roots can express either a stative/quality meaning or an inchoative one (e.g., *nfr*: 'to be/become good') depending on the verbal form. With infinitives, however, both meanings are often possible and difficult to distinguish. For this reason, I will use a single, overarching category, i.e. QUALITATIVE/INCHOATIVE, while recognizing that some verbs in this category may lean more toward a qualitative/stative interpretation, while others may have a more inchoative meaning. Naturally, this category will have limited diagnostic value due to its inherent semantic ambiguity.

I will also exclude the autobenefactive domain from consideration here, as the examples from Cushitic and Semitic demonstrate that the range of verbs potentially carrying an autobenefactive nuance is so broad that it could readily apply to most verbs in the corpus, making it a less effective diagnostic category.

Finally, it is important to acknowledge that while the semantic category of some verbs is relatively clear, in other cases, the decision to assign a verb to one domain or another can be quite challenging and, ultimately, subjective. As such, other scholars might disagree with my classification of certain verbs. However, this is not a major issue: if a pattern truly exists in the data, it should still be recognizable, even in the case of a few ambiguous instances.

With all this in mind, the verbs from the Pyramid Texts of Unas can be semantically classified as shown in Table 1.

These data can be summarized also in proportional terms, as in Table 2.

These results are suggestive in several ways. First, it is worth noting that the corpus used here contains very few verbs that can be assigned to the categories of RECIPROCAL and GROOMING/BODY-RELATED verbs. This is likely, at least in part, a feature of the corpus itself, as verbs such as 'to shave' and 'to comb' are attested elsewhere in Old Egyptian (both are *3-EtR* [–t]). However, the Pyramid Texts are funerary ritual texts, so it is possible that these semantic domains simply fall outside their scope. In the case of RECIPROCAL verbs, the issue might also stem from modern translations: since reciprocity is often a subtle

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TABLE 1 Semantic distribution of verbs in the Pyramid Texts of Unas

		2⁺-EtR [–t]	*2⁺-EtR* [+t]	*3-EtR* [–t]	*3-EtR* [+t]
TOTAL VERBS		**53**	**85**	**173**	**3**
1	NON-T-ASSOCIATED DOMAINS	33	24	115	0
2	QUALITATIVE/INCHOATIVE	11	17	22	0
3	T-ASSOCIATED DOMAINS—TOTAL	9	44	36	3
WITHIN THE T-ASSOCIATED DOMAINS		**9**	**44**	**36**	**3**
A	RECIPROCAL			1	1
B	GROOMING/BODY-RELATED		1		
C	GIVE BIRTH		1		
D	EMOTIONS AND COGNITIVE VERBS	2	6	5	
E	TRANSLATIONAL/POSTURAL MOVEMENT	6	22	16	1
	Basic movements		*3*		
	Movements form a source		*4*		
	Movement through something		*4*		
	Types of movement	*2*	*4*	*4*	
	Change of direction		*2*	*2*	
	Movement toward/into/onto a target	*4*	*3*	*3*	
	Movement around something		*1*	*2*	
	Moving from one place to another			*3*	
	Postural movements		*1*	*2*	*1*
F	MOVEMENT INVOLVING OTHERS	1	14	14	1
	To take/carry	*1*	*5*	*6*	*1*
	To put/give/place		*5*	*1*	
	To send away		*3*	*5*	
	To move apart/separate		*1*	*2*	

nuance, it is possible that verbs perceived as reciprocal by the ancient Egyptians have not been recognized as such by modern Egyptology. A more refined definition of which concepts are typically expressed with reciprocal verbs in Afroasiatic languages, coupled with a target corpus-based study of these concepts in Egyptian texts, might uncover some overlooked semantics. A study of this kind would, however, be a task for future research.

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TABLE 2 Proportional distribution by verbal class. The highest percentage for each class is highlighted in grey

		2⁺-EtR [–t]	*2⁺-EtR* [+t]	*3-EtR* [–t]	*3-EtR* [+t]
TOTAL VERBS		**53**	**85**	**173**	**3**
1	NON-T-ASSOCIATED DOMAINS	62%	28%	66%	0%
2	QUALITATIVE/INCHOATIVE	21%	20%	13%	0%
3	T-ASSOCIATED DOMAINS—TOTAL	17%	52%	21%	100%
WITHIN THE T-ASSOCIATED DOMAINS		**9**	**44**	**36**	**3**
A	RECIPROCAL			3%	33%
B	GROOMING/BODY-RELATED		2%		
C	GIVE BIRTH		2%		
D	EMOTIONS AND COGNITIVE VERBS	22%	14%	14%	
E	TRANSLATIONAL/POSTURAL MOVEMENT	67%	50%	44%	33%
F	MOVEMENT INVOLVING OTHERS	11%	32%	39%	33%

Turning to the distribution of verbs across these classes, it is clear that *3-EtR* [+t] verbs are extremely rare overall; nevertheless, all three of the attested examples fall within domains typically associated with the *t* morpheme in other Afroasiatic languages. In contrast, *3-EtR* [–t] verbs constitute the most common group. It is notable that a majority of these verbs appear in non-t-associated domains (66%), compared with only 21% in t-associated ones. A similar pattern is observed for *2⁺-EtR* [–t] verbs, with 62% occurring in non-t-associated domains and only 17% in t-associated ones.

Against this backdrop, the situation for *2⁺-EtR* [+t] verbs is particularly remarkable. In this group, the pattern is effectively reversed: these verbs are more frequently found in t-associated domains (52%) than in non-t-associated ones (28%). Furthermore, in absolute numbers, *2⁺-EtR* [+t] verbs in t-associated domains (44) outnumber the *3-EtR* [+t] verbs in the same domains (36), even though *3-EtR* [+t] verbs are far more abundant overall (173 versus 85 verbs).

Overall, it appears that verbs whose infinitives are characterized by the presence of a final *-t* are clearly predominatly associated with t-associated domains (*2⁺-EtR* [+t] = 52%; *3-EtR* [+t] = 100%), while verbs whose infinitive lack the -t

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are predominantly found with non-t-associated ones (*2*$^{+}$*-EtR* [–t] = 62%; *3-EtR* [–t] = 66%). These patterns are unlikely to be coincidental and point to a clear correlation between the presence of the *-t* suffix of Egyptian infinitives and *t*-associated semantic domains, providing strong support for the initial hypothesis that the infinitival -t is connected to the Afroasiatic t morpheme.

The fact that *2*$^{+}$*-EtR* [+t] verbs also appear outside the t-associated domains does not undermine the observed correlation. The 17 verbs in quality domains could be interpreted as instances of the uncontrolled inchoative value of the Afroasiatic **t*, while the 24 verbs in other non-t-associated domains may reflect either specific developments within Egyptian semantics or secondary functions of the Afroasiatic t morpheme, such as its autobenefactive value. Notably, a preliminary survey of these verbs reveals parallels in both Semitic and Cushitic languages, where similar verbs, though falling outside the canonical domains considered here, are also occasionally characterized by the t morpheme. A deeper understanding of these semantic fields across Afroasiatic languages, however, would provide a firmer framework for interpreting the Egyptian data.

In this regard, it is noteworthy that *2*$^{+}$*-EtR* [–t] verbs of movement are predominantly associated with movement toward, into, or onto, while largely absent from other movement-related domains. It would be valuable to investigate whether similar patterns occur in other Afroasiatic languages.

Finally, it is also noteworthy that translational and postural movement verbs constitute a major group of t-compatible verbs. The use of the Afroasiatic **t* morpheme with movement verbs has clear parallels in other languages, although the underlying semantic motivation is not fully understood (see e.g., Kouwenberg 2005 for Akkadian). One might, however, easily imagine reflexive nuances, such as 'I move myself', or autobenefactive nuances, such as 'I move for my own sake'. Interestingly, in Egyptian, movement verbs often display some degree of inherent transitivity; for example, Coptic [S] ⲡⲣⲉ |pírə| < Egyptian *pr-t* can mean both 'to come forth [intr.]' and 'to put forth [trans.]'—in the frame presented above, these could be parsed as 'the act of being put forth by oneself' = 'to come forth [intr.]' and 'the act of being put forth by someone' = 'to put forth [trans.]'.

5 Conclusions

In conclusion, this paper has explored three potential cases where the Afroasiatic **t* morpheme might be reflected in Ancient Egyptian verbal morphology. While all three cases were previously proposed by other scholars the present

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study aimed to recontextualize the first two in light of recent scholarship and to offer a completely new analysis for the third. In particular, whereas Loprieno's proposal sought to connect all these forms through a supposed original perfective meaning of the **t* morpheme, the approach developed here, building on more recent research, favors an original middle meaning for the morpheme. Whether the evidence presented is convincing or not is, of course, left to the reader.

I believe, however, that ample room for further investigation remains, especially concerning the third case. This could involve defining a more precise typology of the t morpheme across Afroasiatic languages, which would then provide a stronger basis for assessing the Egyptian data, as well as a more systematic analysis of Egyptian forms and their attestations across different corpora, including the Pyramid Texts of other kings, other Old Egyptian texts beyond the Pyramid Texts, and even Early Middle Egyptian corpora such as the Coffin Texts.

Overall, while Egyptian evidence may offer only limited new insights into the Afroasiatic context in this instance, a deeper understanding of Afroasiatic languages could significantly enhance our comprehension of several features of Egyptian verbal morphology. It is hoped that future research will be able to further support, or convincingly challenge, at least some of the ideas presented here.

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